

EyeMeet: A Joint Attention Support System for Remote Meetings

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ABSTRACT

A major challenge in remote meetings is that awareness cues, such as gaze, become degraded despite playing a crucial role in communication and establishing joint attention. Eye tracking allows overcoming these obstacles by enabling augmentation of remote meetings with gaze information. In this project, we followed a participatory approach by first distributing a scenario-based survey to students ($n=79$) to uncover their preference of eye-based joint attention support (real-time, retrospective, real-time & retrospective, no) for remote university meetings. Building on these findings, we developed EyeMeet, an eye-based joint attention support system that combines state-of-the-art real-time joint attention support with a retrospective attention feedback for remote meetings. In a four-week study, two student groups worked remotely on course assignments using EyeMeet. Our findings of the study highlight that EyeMeets supports students in staying more focused on the meetings. Complementing real-time joint attention support, retrospective joint attention feedback is recognized to provide valuable support for reflecting and adapting behavior for upcoming meetings.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Collaboration, Joint attention, Eye tracking, Remote meeting, Feedback

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1 INTRODUCTION

In times of the ongoing COVID-19 pandemic, education at universities has vastly transformed. Remote lectures, tutorials, and group meetings are becoming the new normal. Allocating attention properly in remote meetings is known as one of the key challenges

people have [13, 18]. Specifically, sustained attention is crucial in a learning context [31]. Besides linguistic, para-linguistic, and awareness cues, such as gestures, facial expressions and gaze direction, play an important role in transmitting messages and establishing joint attention [15]. In particular, awareness cues support collaborators to concurrently allocate attention to the same object. However, awareness cues are missing partially or entirely in remote meetings making it difficult to establish joint attention. Ultimately, this negatively impacts collaboration efficiency and effectiveness [24].

Gaze direction is known as one of the critical channels to increase joint attention. Therefore, eye tracking is a promising technology for supporting joint attention in remote and co-located collaboration scenarios [9]. Eye tracking technology enables augmenting collaboration with real-time and retrospective gaze information of collaborators. Previous studies have shown that real-time eye-based joint attention support increases the efficiency and effectiveness of collaboration in programming, writing, games, search tasks and remote instructions [3, 4, 6, 12, 16, 17, 30]. One well-researched approach is sharing the presenter's eye movements with collaborators in real-time (dual eye tracking) via shared gaze visualizations [9]. An alternative approach is retrospective attention feedback which is usually given through comprehensive metrics and visualizations presented on a dashboard [19, 32]. While prior work primarily focused on real-time gaze sharing to improve joint attention in collaborative meetings, user preferences for different types of joint attention support (real-time vs. retrospective) are currently not well understood. Moreover, to the best of our knowledge, there is no system that integrates both real-time and retrospective support to improve joint attention in remote meetings.

In this paper, we first investigated students' preferences with regards to receiving joint attention support in remote meetings in a scenario-based survey study ($n=79$). The survey results highlight that two thirds of the students in general would like to have joint attention support, be it real-time, a retrospective feedback, or preferably a combination of both as joint attention support. Based on this empirical evidence, we developed a prototypical artifact, called EyeMeet, that integrates real-time and retrospective joint attention support. The goal of EyeMeet is to augment collaboration by helping users to establish, sustain, and reflect joint attention during and after remote video conferences. EyeMeet integrates an eye tracker at each remote student workplace to enable sharing a real-time circular shared gaze visualization of the presenter's gaze in remote meetings. Moreover, EyeMeet provides a complementary individual retrospective joint attention feedback as a dashboard, the so-called AttentionBoard. It depicts attention metrics to reflect attention distribution during the last meeting and to support students to increase awareness about their joint attention status. We conducted a four-week study with two teams of students ($n=6$) using EyeMeet while working remotely on course assignments in

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weekly Microsoft Teams meetings. The students had the option to activate the real-time shared gaze visualization during the meeting and received the AttentionBoard as retrospective attention feedback. We gathered users' insights about their experiences using EyeMeet by conducting interviews as well as collected quantitative data of their behavior in remote meetings.

Our results highlight that students want to have both types of joint attention support features for remote meetings. Based on the students' experiences during the study, we found that the real-time shared gaze feature supported them to focus on the meeting, and they experienced habituation to the gaze visualization. Moreover, the retrospective AttentionBoard helped students to reflect their attention distribution and on this basis adapt their behavior accordingly in upcoming meetings. These insights are also backed by the quantitative data showing that EyeMeet leads to a higher joint attention of students' during remote meetings.

2 RELATED WORK

Being aware of the involved collaborators is a well known challenge in remote collaboration [8]. Research has taken various approaches to overcome the threat of missing awareness cues in remote meetings, such as gestures, body movements, facial expressions, and eye movements. Eye gaze has been known as a valuable resource to improve communication and collaboration of humans [34]. Joint visual attention of collaborators describes that two or more people concurrently allocate attention to the same object and it is critical in collaboration. Over the last years, scholars have applied eye tracking technology to support collaboration in different tasks and contexts typically by providing shared gaze visualization in real-time [9]. Shared gaze visualizations represent the user's gaze direction on an user interface to make the point of gaze visible for remote collaborators and increasing awareness. Previous studies testing several shared gaze visualizations have shown that this real-time feature leads to a higher joint visual attention, better quality of collaboration, lower task completion times and higher accuracy [9]. In collaborative search tasks, for example, shared gaze visualizations have positively affected collaborators' performance, coordination, and searching behavior [3, 3, 6, 22, 30, 34]. Furthermore, gaze sharing has been investigated on remote collaboration tasks such as joint programming projects [4], writing academic texts [16], problem-solving [5, 21] and remote instructions [1, 12, 14]. In a learning context, shared gaze visualizations have been investigated to support students and to facilitate communication [9]. For example, it has been shown that learners could more easily follow the presenter, experienced a higher quality of collaboration and learning gain when seeing the collaborator's gaze position [23, 27, 29].

Besides, research has also shown that retrospective feedback can be effective for positively influencing behavior of individuals. Typically, retrospective feedback is provided in the form of a dashboard [28]. A dashboard visualizes aggregated data on single screen in a comprehensive way [11]. Dashboards have proven to be supportive in several collaboration and also learning scenarios [10, 20, 25, 26, 33]. Samrose et al. [25] have investigated the design of a personalized dashboard MeetingCoach that includes behavioral meeting information about engagement, tonality, speaking

patterns and a meeting video including a transcript. MeetingCoach improved users' awareness of meeting dynamics and affected inclusivity and effectiveness. Ez-zaouia et al. [10] have designed the dashboard EMODASH that helps tutors to increase their awareness of learner's emotions and ultimately tutors incorporated more affective elements in their written feedback.

Previous work has mainly focused on real-time joint attention support in controlled environments. EyeMeet is the first attempt to combine state-of-the-art real-time and retrospective joint attention support for remote meetings. Moreover, our work collects empirical data about student preferences and outcomes of using a joint attention support artifact in the field.

3 SCENARIO-BASED SURVEY STUDY

As the first step, our goal was to get a deeper understanding of what type of joint attention support systems (real-time, retrospective, real-time and retrospective, no) students would prefer for their remote meetings at university. We specifically focused on remote meetings in which students work on group assignments. Because of its different nature, traditional lectures were out of scope in our scenario-based survey study. In the first part of the survey, eye tracking technology was introduced through a video providing information about the technology and its capabilities. Afterwards both real-time and retrospective joint attention supports were introduced in a random order via a video. The video demonstrated in detail the usage of both supports during and after a remote meeting. Participants articulated the perceived usefulness on a 7 point Likert-Scale [7]. Ultimately, participants stated their preference of joint attention support systems by ranking it from the most preferred to the least preferred. The survey was distributed to students through a university experimental panel and participating students received 5€. In total 79 students (33 females, 46 males) completed the survey and their average age was 24,04 years ($SD=2.71$ years). Participants had no to little previous experience with eye tracking technology on average ($M=1.38$, $SD=0.61$).

3.1 Key Findings

The survey responses highlighted that participants did not perceive the real-time and retrospective joint attention supports differently in terms of usefulness (real-time: $M=4.22$, $SD=1.75$; retrospective: $M=4.26$, $SD=1.61$). Figure 1 shows a Sankey diagram with the number of participants that ordered their preferences the same way. All-in-all, twice as many people most preferred to receive some kind of joint attention support compared to no joint attention support. Two thirds of the survey participants said that they prefer either the real-time, the retrospective or the combination of both joint attention supports the most. Participants that would like to have both joint attention support systems preferred the real-time support more than the retrospective support (see red path). One third said that they would rather prefer not to receive a joint attention support. Overall, the majority of these participants would prefer the retrospective over the real-time or both joint attention supports (see purple path).

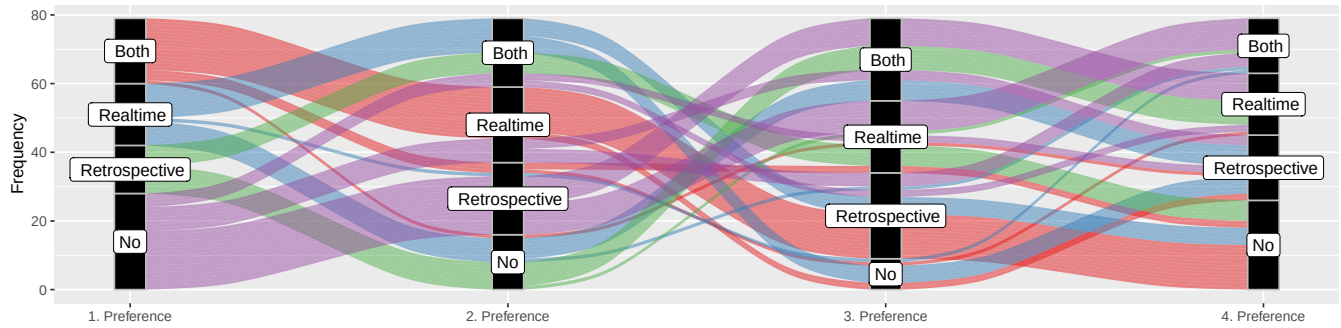


Figure 1: Ranking of Joint Attention Support System Configuration Preferences

4 THE EYEMEET SYSTEM

The goal of EyeMeet is to augment collaboration by supporting students in establishing, sustaining, and reflecting joint attention during and after remote team meetings. Augmentation of collaboration is achieved by leveraging capabilities of eye tracking technology and enriching remote collaboration by gaze information in real-time and retrospective. As the real-time joint attention support, we use the circular shared gaze visualization through which presenter can visualize eye movements while sharing their screen in recurring meetings. The presenter is typically the one involved in the meeting by sharing some content via the screen sharing function of the video conferencing tool. As the retrospective attention feedback, we design a personalized dashboard, called AttentionBoard, that summarizes the joint attention status during the last meeting.

4.1 Real-time Shared Gaze Visualization

In this study, the Tobii Eye Tracker 4C was used, which comes with the Tobii Core Software. The Tobii Core software integrates the gaze trace function which provides a circular overlay of the gaze position on the user interface. The gaze trace function is integrated into EyeMeet to present the real-time shared gaze visualization. When presenting and sharing the screen during a Microsoft Teams meeting, the presenter activates the circular shared gaze visualization in EyeMeet upon demand. Only the presenter's gaze visualization can be visible at a specific point in time. If another meeting attendee wants to share his/her gaze, then this person has to become the presenter. By transferring the shared gaze visualization with the screen sharing function, attendees can follow the presenter's focus of attention easily and the joint attention of attendees and presenter is supposed to increase. When changing the roles, the presenter can turn off the gaze trace function in EyeMeet, and the new presenter can turn it on again. In Figure 2 the shared gaze visualization is depicted during a Microsoft Teams online meeting.

4.2 AttentionBoard

The AttentionBoard provides individualized retrospective feedback about attention distribution and joint attention with presenters during remote meetings. It aims to enable students to reflect their attention distribution after the group meeting and adapt their behavior based on provided information in the next meeting. To design AttentionBoard, the eye movement data of all participants

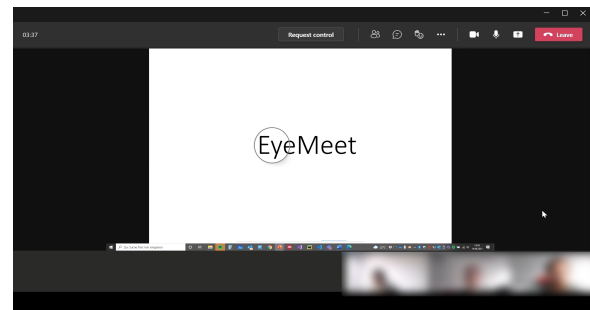


Figure 2: Circular Shared Gaze Visualization of EyeMeet Shared via Microsoft Teams

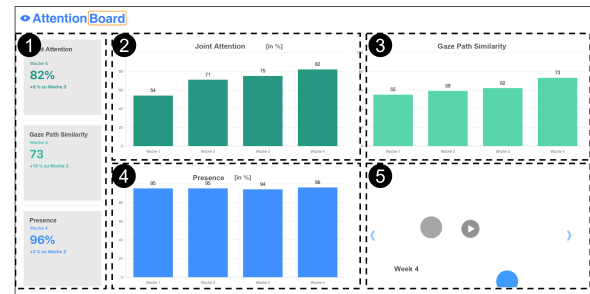


Figure 3: Design of AttentionBoard and its components.

are recorded during the meeting. AttentionBoard only provides the students's joint attention key performance indicators (KPIs) for the time that students were not in the role of the presenter. AttentionBoard is depicted in Figure 3.

On the left side of AttentionBoard (1), an overview of all metrics of the last meeting is presented in a text format. On the right side, the joint attention KPIs are provided as bar graphs. Percentage of Joint Attention (JA) (2) is defined as the proportion of the time the student as a listener spent looking on the same area as the presenter [4, 17]. It is calculated based on the gaze overlap metric [3, 4, 6, 17, 27]. This KPI provides insights into how well the user paid attention to what the presenters showed during the meeting. Fixations of listeners and presenter were calculated for each part

of the meeting. Based on these fixations, the software checks for every second whether there was joint attention.

The gaze-path similarity (GS) index (3) indicates how well the students followed content presentations during the meeting. This metric is based on scan path analysis of eye movement data. It leverages the Levenshtein distance to calculate the similarity of the student's and presenter's scan paths by converting them to strings [2]. To calculate GS, a grid is laid over the screen, and each cell receives a unique identifier. Fixations are then translated to a string based on the identifier of the cell that the fixation occurred. These two strings are taken to calculate the Levenshtein distance. Both metrics, JA and GS, were calculated for each participant_i and meeting_j as the weighted average of the JA_{ijk} or GS_{ijk} during each section_k of the corresponding meeting_j. Listener_{ijk} equals to 1 when the participant_i was a listener in the corresponding section_k or to 0 if not.

$$JA_{ij}/GS_{ij} = \frac{\sum_{k=1}^n JA_{ijk}/GS_{ijk} \cdot Duration_{jk} \cdot Listener_{ijk}}{\sum_{k=1}^n Duration_{jk} \cdot Listener_{ijk}} \cdot 100\% \quad (1)$$

The user's presence during the meeting is shown on the AttentionBoard as well (4). This metric leverages the recorded eye tracking data to calculate the percentage of time the user was paying attention to the meeting. AttentionBoard also provides a video about the user's attention distribution during the remote meeting (5). This is a new type of attention data visualization for reflecting meetings. It depicts the student's gaze point (blue circle) and the presenter (grey circle) on a neutral white background. Figure 4 shows the possible frames of the video. When circles are overlapping, it means that the student had joint attention with the presenter (1). In contrast, no joint attention was reached when circles are not overlapping (2). Frames that contain only one blue (3) or one grey circle (4) indicate that only the listener or the presenter was looking at the display in that specific second.

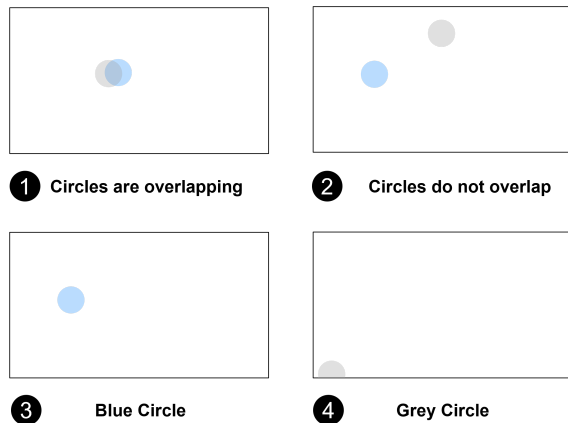


Figure 4: Frames of Video on AttentionBoard

5 EVALUATION STUDY

A evaluation study was designed to investigate the usage of the EyeMeet artifact and explore its effects on joint attention of meeting participants. The study design went through the institutional review process in terms of data security and was approved. During a four-week field study the EyeMeet artifact was used by students in their weekly remote meetings. In total, two groups of students (two females, one male per group) from the university experimental panel participated on a voluntarily basis in this study. On average, participants were 21.83 years ($SD = 2.64$ years) old. None of the participants had previous experience with eye tracking technology. In Group1 (participants P1-P3) students accomplished a project which was part of a lecture. Group2 (participants P4-P6) consisted of students working on weekly assignments of their study program. Participants were compensated with 84€.

Before the study, we provided each participant a Tobii Eye Tracker 4C with the license for research. Participating students used their own computers to participate in the study from home. Monitor resolutions varied from 1366x788 pixels to 3000x2000 pixels. Two participants had a dual monitor setup during the experiment with the eye tracker being connected to the main monitor, which also displayed the Microsoft Teams window. During the meetings, the Microsoft Teams window was always supposed to be on full-size when being the active window.

5.1 Procedure

Before the experiment, we conducted a 1.5 hours introduction workshop with each group via Microsoft Teams to introduce the study to the participants. In this workshop, we briefed them about eye tracking technology, the study background, and the study's procedure. Moreover, the eye tracker and experimental software were set up together with the experimenter step by step remotely. At the end of the workshop, each participant tested the eye tracker and software together with the experimenter to validate the correctness of the setup for the study. Figure 5 shows the study's procedure.

In total, we collected data of the students' weekly remote meetings in Microsoft Teams over four weeks. The first week served as the control condition and no joint attention support was provided to the students. From Week2 on, presenters were allowed to use the circular shared gaze visualization of EyeMeet while sharing their screen to present content. In all meetings, eye movement and log data were recorded by the EyeMeet software from all participants in the background. The recorded data was used to calculate the metrics and visualizations for the individualized AttentionBoard. The remote meetings were at least 30 minutes and a maximum of 1.5 hours long. In each meeting, participants worked together on their assignment, and at least two participants presented content to the team members by sharing their screen. Participants could switch between presenter and listener roles during the meeting by starting and stopping sharing their screens. At the end of each week, participants received a notification that an updated version of AttentionBoard with data from the last meeting is ready in order to reflect the last meeting.

Over the course of the field study, both groups faced challenges in data collection in at least one meeting, and data of these meetings had to be excluded from the analysis. For each excluded meeting of

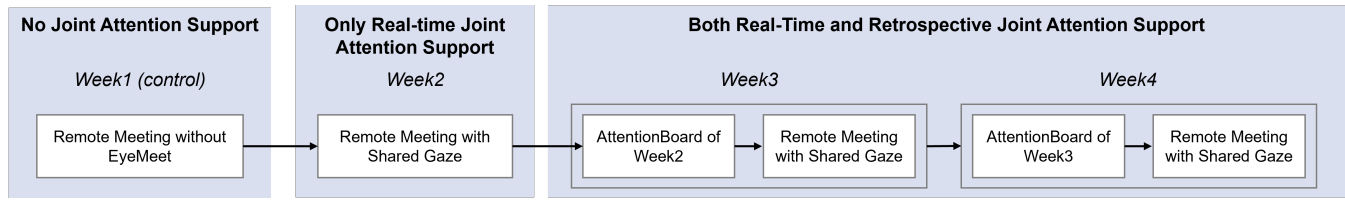


Figure 5: Study Procedure

Group1 and Group2, the study was extended by another week for the corresponding group so in total valid data of four weeks was recorded (two additional weeks for Group1, one additional week for Group2).

5.2 Results

At the end of the study, each participant was interviewed for 30 minutes. Three interviewees described the shared gaze feature as useful, particularly when presenting complex mathematical formulas or packed slides of a presentation. P2 reported *"I found the gaze bubble feature especially cool because you really knew where the other person was looking at and where you should focus your attention"*. After using the circular shared gaze visualization for several weeks, 33% of participants experienced habituation to the feature and it lost its eventually disturbing character. Two-thirds of the participants experienced that the shared gaze visualization increased their focus on the meeting and presentations and helped them ignore external distractions like smartphones when working on their project. Additionally, two participants stated that their meeting productivity increased when using the shared gaze feature. P2 remarked *"I would definitely say that the output increased. It's not that you just pretend to be attentive"*. Two interviewees would like to continue to use the shared gaze feature in the future. When inspecting AttentionBoard, five of six interviewees reported that they reflected their behavior during the last remote meeting. Moreover, three out of six participants stated that they tried to adapt their behavior accordingly in the next meeting. P1 commented *"When I had these low values for joint attention, I tried to be more attentive"*. Adding to that P4 and P5 asked to have a feature that allows them to perform a deep dive into the metrics, e.g. a drill-down functionality to reflect the meeting section by section.

We also evaluated the EyeMeet system and its effects on the behavior quantitatively by the two metrics that we also used to design AttentionBoard as described in chapter 4: percentage of time with JA and GS.

Figure 6 presents the averages and a boxplot of the JA for each week. It can be seen that the JA increased each week of the recurring meetings from an average of 43% ($SD = 7.90\%$) in Week1 (control condition) to 65% JA ($SD = 8.00\%$) in Week4. The results from Friedman's ANOVA test demonstrated that the mean JA significantly changed over the four weeks of the experiment, $\chi^2(3) = 9.60, p = .022$. Conducting post-hoc tests using Bonferroni correction, $p_{adj.}$ -values are above the required level of 0.05 for the comparison of Week1-Week2, $p_{adj.} = 1, r = .16$, Week1-Week3, $p_{adj.} = .19, r = .44$, and Week1-Week4, $p_{adj.} = .19, r = .44$. Therefore, in none of the weeks JA was significantly higher than in Week1

which can be due to the small sample size of $n=6$. Altogether, the post-hoc tests indicate a trend of improvement in JA for Week3 and Week4, supported by medium effect size, $r > .3$.

Figure 7 shows that the mean GS Index increased just when both joint attention supports were available in Week3 ($M = 70.6, SD = 4.62$) and in Week4 ($M = 71.1, SD = 2.89$). The results from Friedman's ANOVA test show that the mean GS significantly changed over the four weeks of the experiment, $\chi^2(3) = 11, p = .012$. Post-hoc tests using Bonferroni correction show that the differences between weeks are not significant, Week1-Week2, $p_{adj.} = .1, r = .12$, Week1-Week3, $p_{adj.} = .19, r = .44$, and Week1-Week4, $p_{adj.} = .38, r = .38$. However, medium effect sizes ($r > .3$) demonstrate that the EyeMeet software with the gaze sharing feature and AttentionBoard tends to increase GS in Week3 and Week4.

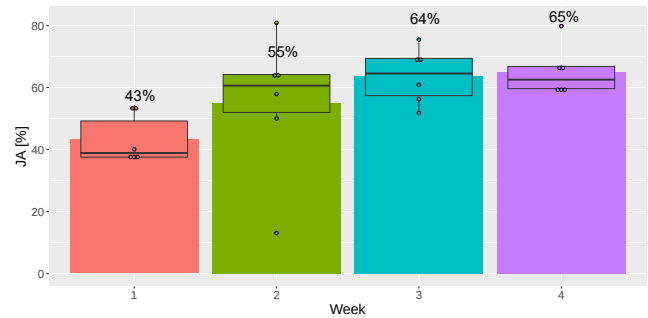


Figure 6: Boxplot of JA for Each Week

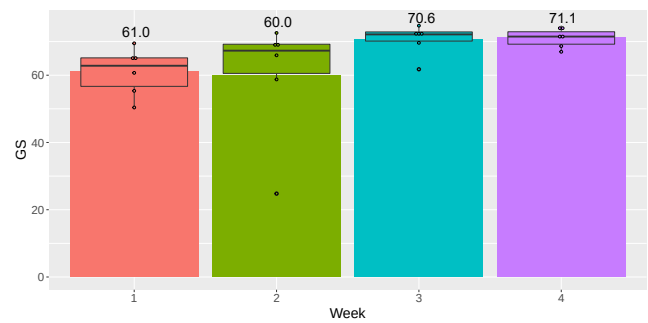


Figure 7: Boxplot of GS for Each Week

6 DISCUSSION & FUTURE WORK

We first gathered students preference for joint attention support in remote meetings in a scenario-based survey (n=79). Based on the survey result, we designed EyeMeet, a novel artifact that combines a state-of-the-art real-time with a new retrospective support to improve joint attention during remote meetings. The retrospective joint attention dashboard was developed to enable students to reflect their joint attention status during meetings. We evaluated EyeMeet in a four-week study to gather insights about student's behavior and usage of joint attention support in the field.

All in all, our results suggest that students would like to receive an eye-based joint attention support in remote group meetings at university. Moreover, students derive value from the EyeMeet's real-time and retrospective joint attention support as it enables them to stay more focused on the meeting and increases awareness about the joint attention status during remote meetings to adapt it in upcoming meetings. These insights are also supported by the quantitative results of joint attention and gaze paths similarity metrics showing an increase when using EyeMeet. A limitation of these findings is the small study sample size of six participants.

Joint attention is essential for effective and efficient communication and collaboration, especially in remote settings. The quantitative results of this study suggest that maximum joint attention stagnates at a certain level. This raises the questions if maximum joint attention is the ultimate goal to improve communication and collaboration, and what is the best level of joint attention to reach maximum efficiency and effectiveness. The evolvement of joint attention over the four weeks of the study also rises the question about long-term effects of joint attention support and learning effects. Moreover, this study focused on supporting joint attention during presentations in remote meetings and did not cover further scenarios like conversations, discussions or negotiations. Furthermore, there is a potential of integrating gaze visualizations of the audience in remote meetings as well. To answer these questions, there is a need for further research.

In terms of retrospective joint attention feedback, this study proposes a first design which comes with some shortcomings and opportunities for further improvement. Participants articulated the need to be able to further deep dive into the metrics of AttentionBoard as the current metrics are high level. Providing the metrics at a finer level, like per meeting session segment, would allow users to further drill-down in which parts of the meeting they were inattentive. In future, there exists also the possibility to provide the metrics in (near) real-time to the user.

REFERENCES

- [1] Deepak Akkil, Biju Thankachan, and Poika Isokoski. 2018. I see what you see: Gaze awareness in mobile video collaboration. In *Eye Tracking Research and Applications Symposium (ETRA)*. Association for Computing Machinery, New York, NY, USA, 9. <https://doi.org/10.1145/3204493.3204542>
- [2] Stephan A. Brandt and Lawrence W. Stark. 1997. Spontaneous eye movements during visual imagery reflect the content of the visual scene. *Journal of Cognitive Neuroscience* 9, 1 (1997), 27–38.
- [3] Susan E Brennan, Xin Chen, Christopher A Dickinson, Mark B Neider, and Gregory J Zelinsky. 2008. Coordinating cognition : The costs and benefits of shared gaze during collaborative search. *Cognition* 106 (2008), 1465–1477.
- [4] Sarah D'Angelo and Andrew Begel. 2017. Improving Communication Between Pair Programmers Using Shared Gaze Awareness. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 6245–6255.
- [5] Sarah D'Angelo and Darren Gergle. 2016. Gazed and Confused : Understanding and Designing Shared Gaze for Remote Collaboration. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 2492–2496.
- [6] Sarah D'Angelo and Darren Gergle. 2018. An Eye For Design: Gaze Visualizations for Remote Collaborative Work. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [7] Fred D. Davis. 1989. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly* 13, 3 (1989), 319–340.
- [8] Paul Dourish and Victoria Bellotti. 1992. Awareness and coordination in shared workspaces. In *Proceedings of the Conference on Computer-Supported Cooperative Work*. Association for Computing Machinery, New York, NY, USA, 107–114. <https://doi.org/10.1145/143457.143468>
- [9] Sarah D'Angelo and Bertrand Schneider. 2021. Shared Gaze Visualizations in Collaborative Interactions: Past, Present and Future. *Interacting with Computers* 00, 0 (2021), 19. <https://doi.org/10.1093/iwcomp/iwab015>
- [10] Mohamed Ez-zaouia, Aurélien Tabard, and Elise Lavoué. 2020. EMOASH: A dashboard supporting retrospective awareness of emotions in online learning. *International Journal of Human Computer Studies* 139, February (2020), 102411. <https://doi.org/10.1016/j.ijhcs.2020.102411>
- [11] Stephen Few. 2006. *Information Dashboard Design*. O'Reilly, Sebastopol, CA.
- [12] Kunal Gupta, Gun A. Lee, and Mark Billinghurst. 2016. Do you see what i see? the effect of gaze tracking on task space remote collaboration. *IEEE Transactions on Visualization and Computer Graphics* 22, 11 (2016), 2413–2422.
- [13] Carl Gutwin and Saul Greenberg. 2002. A descriptive framework of workspace awareness for real-time groupware. *Computer Supported Cooperative Work* 11, 3-4 (2002), 411–446. <https://doi.org/10.1023/A:1021271517844>
- [14] Keita Higuchi, Ryo Yonetani, and Yoichi Sato. 2016. Can Eye Help You?. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 5180–5190.
- [15] Robert E Kraut, Susan R Fussell, Susan E Brennan, and Jane Siegel. 1990. Understanding Effects of Proximity on Collaboration: Implications for Technologies to Support Remote Collaborative Work. In *Distributed Work*. MIT Press, Cambridge, MA, USA, 137–162.
- [16] Grete H.; Kütt, Kevin; Lee, Ethan; Hardacre, and Alexandra; Papoutsaki. 2019. Eye-Write: Gaze Sharing for Collaborative Writing. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [17] Grete H. Kütt, Teerapaun Tanprasert, Jay Rodolitz, Bernardo Moyza, Samuel So, Georgia Kenderova, and Alexandra Papoutsaki. 2020. Effects of Shared Gaze on Audio- Versus Text-Based Remote Collaborations. *Proceedings ACM Human-Computer Interaction* 4, CSCW2 (10 2020), 25.
- [18] Anastasia Kuzminykh and Sean Rintel. 2020. Classification of Functional Attention in Video Meetings. In *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376546>
- [19] Moritz Langner, Peyman Toreini, and Alexander Maedche. 2020. AttentionBoard: A Quantified-Self Dashboard for Enhancing Attention Management with Eye-Tracking. In *Information Systems and Neuroscience. NeuroIS 2020. Lecture Notes in Information Systems and Organisation*. Springer, Cham, 266–275. http://link.springer.com/10.1007/978-3-030-60073-0_31
- [20] Gilly Leshed, Diego Perez, Jeffrey T. Hancock, Dan Cosley, Jeremy Birnholtz, Soyoun Lee, Poppy L. McLeod, and Neri Gay. 2009. Visualizing real-time language-based feedback on teamwork behavior in computer-mediated groups. In *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery, New York, NY, USA, 537–546. <https://doi.org/10.1145/1518701.1518784>
- [21] Romy Müller, Jens R. Helmer, Sebastian Pannasch, and Boris M. Velichkovsky. 2012. Gaze transfer in remote cooperation: Is it always helpful to see what your partner is attending to? *Quarterly Journal of Experimental Psychology* 66, 7 (2012), 1302–1316.
- [22] Mark B. Neider, Xin Chen, Christopher A. Dickinson, Susan E. Brennan, and Gre Gory J. Zelinsky. 2010. Coordinating spatial referencing using shared gaze. *Psychonomic Bulletin and Review* 17, 5 (2010), 718–724.
- [23] Pernilla Qvarfordt and Matthew Lee. 2018. Gaze patterns during remote presentations while listening and speaking. In *Eye Tracking Research and Applications Symposium (ETRA)*. Association for Computing Machinery, New York, NY, USA, 9. <https://doi.org/10.1145/3204493.3204540>
- [24] Daniel C. Richardson and Rick Dale. 2005. Looking To Understand: The Coupling Between Speakers' and Listeners' Eye Movements and Its Relationship to Discourse Comprehension. *Cognitive Science* 29, 6 (2005), 1045–1060.
- [25] Samiha Samrose, Daniel McDuff, Robert Sim, Jina Suh, Kael Rowan, Javier Hernandez, Kevin Moynihan, and Mary Czerwinski. 2021. MeetingCoach : An Intelligent Dashboard for Supporting Effective & Inclusive Meetings. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13.

- [26] Samiha Samrose, Ru Zhao, Jeffery White, Vivian Li, Luis Nova, Yichen Lu, Mohammad Rafayet Ali, and Mohammed Ehsan Hoque. 2018. CoCo: Collaboration Coach for Understanding Team Dynamics during Video Conferencing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 4 (2018), 1–24. <https://doi.org/10.1145/3161186>
- [27] Bertrand Schneider and Roy Pea. 2013. Real-time mutual gaze perception enhances collaborative learning and collaboration quality. *International Journal of Computer-Supported Collaborative Learning* 8, 4 (2013), 375–397.
- [28] Beat A. Schwendimann, Maria Jesus Rodriguez-Triana, Andrii Vozniuk, Luis P. Prieto, Mina Shirvani Boroujeni, Adrian Holzer, Denis Gillet, and Pierre Dillenbourg. 2017. Perceiving learning at a glance: A systematic literature review of learning dashboard research. *IEEE Transactions on Learning Technologies* 10, 1 (2017), 30–41. <https://doi.org/10.1109/TLT.2016.2599522>
- [29] Kshitij Sharma, Sarah D'Angelo, Darren Gergle, and Pierre Dillenbourg. 2016. Visual augmentation of deictic gestures in MOOC videos. *Proceedings of International Conference of the Learning Sciences, ICLS* 1, 2012 (2016), 202–209.
- [30] Harri Siirtola, Oleg Špakov, Howell Istance, and Kari Jouko Räihä. 2019. Shared Gaze in Collaborative Visual Search. *International Journal of Human-Computer Interaction* 35, 18 (2019), 1693–1705.
- [31] Ricarda Steinmayr, Mattias Ziegler, and Birgit Träuble. 2010. Do intelligence and sustained attention interact in predicting academic achievement? *Learning and Individual Differences* 20, 1 (2010), 14–18. <https://doi.org/10.1016/j.lindif.2009.10.009>
- [32] Peyman Toreini, Moritz Langner, Alexander Maedche, Stefan Morana, and Tobias Vogel. 2022. Designing Attentive Information Dashboards. *Journal of the Association for Information Systems* 23, 2 (2022), Forthcoming.
- [33] Katrien Verbert, Sten Govaerts, Erik Duval, Jose Luis Santos, Frans Van Assche, Gonzalo Parra, and Joris Klerkx. 2014. Learning dashboards: An overview and future research opportunities. *Personal and Ubiquitous Computing* 18, 6 (2014), 1499–1514.
- [34] Yanxia Zhang, Ken Pfeuffer, Ming Ki Chong, Jason Alexander, Andreas Bulling, and Hans Gellersen. 2017. Look together: using gaze for assisting co-located collaborative search. *Personal and Ubiquitous Computing* 21, 1 (2017), 173–186.